

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce

Department of Information Technology (Bsc.IT Semester III)

Python for Data Science Practical

Practical - II

Roll No. : S036	Name: HARSHAL SHIRKE
Class : SYIT	Batch : 1
Date of Assignment : 22-07-2026	Date/Time of Submission : 22-07-2026

Questions:

2a. Write a python code to create nested tuples.

Code :

```
File Edit Format Run Options Window Help
student = ("HARSHAL",36,("BscIT","semester III"),90)
print("Name =",student[0])
print("Roll No = =",student[1])
print("Department =",student[2][0])
print("Semester =",student[2][1])
print("percentage =",student[-1])
```

Output :

```
===== RESTART: C:/Users/mvlui/Documents/SYIT_
Name = HARSHAL
Roll No = = 36
Department = BscIT
Semester = semester III
percentage = 90
```

2b. Write a python code to sort the nested tuple using the sorted () function.

Code :

```
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```

```
tup = ((1,3), (2,5), (4,2), (5,3))
```

```
Sorted_data = sorted(tup, key = lambda x: x[1])
```

```
print(Sorted_data)
```

```
print("HARSHAL____So36")
```

Output :

```
===== RESTART: C:/Users/mvlui/Documents/SYIT_So
```

```
[(4, 2), (1, 3), (5, 3), (2, 5)]
```

```
HARSHAL____So36
```

2c. Write a python code to copy or clone a list.

Code :

```
tup1 = ((1,3), (2,5), (4,2), (5,3))
```

```
tup2 = ()
```

```
tup2 = tup1
```

```
print(tup2)
```

```
print("HARSHAL____So36")
```

Output :

```
===== RESTART: C:/Users/mvlui/Documents/SYIT_So36/I
```

```
((1, 3), (2, 5), (4, 2), (5, 3))
```

```
HARSHAL____So36
```

2d. Write a python code to check the immutability property of python tuples.

Code :

```
#####  
tup1 = ((1,3), (2,5), (4,2), (5,3))  
tup1[1] = (4,3)  
  
print(tup2)
```

Output :

```
===== RESTART: C:/Users/mvlui/Documents/SYIT_So36/PDS Practical NO2D.py =====  
Traceback (most recent call last):  
  File "C:/Users/mvlui/Documents/SYIT_So36/PDS Practical NO2D.py", line 2, in <module>  
    tup1[1] = (4,3)  
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple**
- b) A tuple with integers only**
- c) A tuple with mixed data types**
- d) A nested tuple (tuple inside another tuple)**

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple**
- Maximum and minimum value (if all are numbers)**
- Count of a specific item**

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
```

```
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- **Find average marks**
- **Display marks above 80**